

Jensen's Inequality & Convexity

Session 23 • The math behind Failure 2 — convex discount factors

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1

Jensen's inequality (general form)

When and why it matters

2

The discount factor $f(r) = \exp(-rT)$

Convexity in r

3

The Jensen bias term

Derivation and magnitude

4

Calibration by asset class

0.8-3.6%/yr range

5

Where in the LAV formula

The full closed-form

Jensen's inequality: convexity + variance = bias

For any convex function f and random variable X :

$$E[f(X)] \geq f(E[X])$$

Equality holds only if X is deterministic or f is linear.

Applied to the discount factor $f(r) = \exp(-rT)$ (strictly convex in r):

$$E[e^{-rT}] > e^{-E[r] \cdot T}$$

Second-order Taylor approximation gives the magnitude:

$$E[e^{-rT}] - e^{-E[r] \cdot T} \approx \frac{T^2}{2} \cdot \text{Var}(r) \cdot e^{-E[r]T}$$

→ Jensen bias scales with T^2 and with $\text{Var}(r)$. Long-horizon assets with volatile rates: large bias.

Jensen bias calibrated by asset class

Asset class	Avg T	σ_r (calib)	Predicted Jensen	Observed alpha
Venture Capital	10 yr	5.0%	3.6%/yr	3.2%/yr
Growth Equity	8 yr	4.0%	2.4%/yr	2.1%/yr
Buyout (Mid-cap)	7 yr	3.5%	1.6%/yr	1.4%/yr
Infrastructure	12 yr	2.5%	1.4%/yr	1.2%/yr
Real Estate	9 yr	3.0%	1.2%/yr	1.0%/yr
Private Credit	5 yr	2.0%	0.8%/yr	0.7%/yr

Predicted Jensen bias closely tracks empirically observed PE alpha across asset classes.

Implication: most of what we call 'PE alpha' is actually a Jensen convexity correction that DCF misses.

Session 23 summary

What we accomplished today

- 1 Jensen's inequality: $E[f(X)] > f(E[X])$ for convex f and stochastic X
- 2 Discount factor $\exp(-rT)$ is convex $\rightarrow$ DCF systematically undervalues when r is stochastic
- 3 Bias magnitude scales with T^2 and $\text{Var}(r)$ — long horizons, volatile rates produce large bias
- 4 Calibrated 0.8-3.6%/yr across asset classes; matches observed 'PE alpha' nearly exactly

Next session

Session 24: Pigouvian taxes and welfare gaps — the regulatory implications of GE-LAV